Venkata Lakshmi Parimala Pasupuleti

AI Engineer · Agentic Runtime · LLM Orchestration · RAG & Memory Retrieval · Multi-Provider LLM Routing · Agent Evaluation · Python · LangGraph

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PROFESSIONAL SUMMARY

AI Engineer who builds the agents themselves — not the backend infrastructure around them, not the UI surfaces on top of them, but the orchestration runtime, reasoning chains, memory retrieval systems, tool integrations, and evaluation frameworks that turn raw LLM capability into production behavior. AgentProbe is a homegrown agent evaluation and observability platform: LLM-as-judge scoring, CI regression harness, multi-provider LLM routing across OpenAI/Anthropic/Groq/Ollama with latency and cost controls, golden datasets, pass@k harness. Leasel is an 8-agent LangGraph system with durable state, hybrid RAG, tool orchestration, and conditional HITL escalation — 93.6% precision across 1,240 test cases. 8+ years of production Python engineering behind both. Uses Cursor and Claude Code as a core part of the daily workflow, not as an afterthought. M.S. Data Analytics Engineering, George Mason University (GPA 3.87). — github.com/venkatalakshmiparimala/AgentProbe

WORK EXPERIENCE

Blue Horse Digital Sep 2026 – Present

AI/ML Engineer (Volunteer) · Remote

- Building ML models and AI-integrated data pipelines; integrating LLM components across backend APIs and reporting layers; debugging reliability issues in production AI-driven features.

Magizhs Tech Solution May 2025 – Aug 2025

AI/ML Engineer Intern · Remote

- Designed and built the full agentic runtime: LangGraph stateful graph (parse → plan → retrieve → validate → conditional route), RAG layer over regulatory corpus with cited-source answers, tool integrations for structured output generation, and validation guardrails as hard enforcement logic. Evaluated emerging LLM models for routing decisions across the pipeline.

George Mason University Aug 2024 – Aug 2025

Research Associate + Teaching Assistant

- Built multi-source prediction pipelines (Python, R) across 190+ countries (R^2=0.977); HuggingFace fine-tuning + inference API deployment on Quart (async); hypothesis testing and model calibration. Workshops to 600+ students across Python and SQL.

Entrata (PropTech / Real Estate SaaS) Sep 2018 – Aug 2024

Senior Software Engineer · Remote

- **Production Python engineering at scale** — owned 20+ backend modules (Python, PostgreSQL, RabbitMQ, AWS) serving 100K+ monthly active users at 99.5% uptime across 120+ Agile sprints. This is the production code ownership and distributed systems foundation behind the AI systems built on top of it.
- **Privacy-preserving systems and compliance** — 100% PHI compliance (Libsodium-encrypted pipelines), HIPAA, SOC 2 across all resident data workflows. Zero security incidents across 7 years. The same privacy-preserving AI design principles that apply to always-on contextual AI on a wearable.
- **Distributed systems, fault tolerance, and observability** — event-driven architecture (RabbitMQ) with idempotency guarantees; multi-tenant PostgreSQL; 200+ query optimizations; Docker microservices (50% scalability improvement); OpenTelemetry + Prometheus + Grafana (40% debugging time reduction). Mentored 5+ engineers; 300+ code reviews.

PROJECTS

AgentProbe: Agent Evaluation & Observability Platform LLM-AS-JUDGE CI-INTEGRATED MULTI-PROVIDER ROUTING

Python · FastAPI · LangGraph · ChromaDB · GitHub Actions · OpenTelemetry | github.com/venkatalakshmiparimala/AgentProbe

- Homegrown agent evaluation and observability framework: 15-endpoint FastAPI backend; 5-metric eval engine (correctness, groundedness, latency, token efficiency, hallucination rate); **LLM-as-judge classifiers** with configurable rubrics; **CI regression harness** with golden datasets and pass/fail gating so agent quality can't silently degrade; **multi-provider LLM routing** across OpenAI/Anthropic/Groq/Ollama with latency and cost controls; offline + online evaluation loops; trace-level observability. This is the evaluation and observability infrastructure for agentic systems — exactly what Awear's platform will need.

Leasel: 8-Agent Orchestration Runtime with RAG & Memory LANGGRAPH HYBRID RAG

Python · LangGraph · FastAPI · ChromaDB · TF-IDF · OCR · SQLite · OpenAI API · LangSmith | github.com/venkatalakshmiparimala/Lease_complaince_rag

- Full agentic runtime: **orchestration layer** managing agent sessions, chaining reasoning steps (Triage → Retrieval → Configuration → Historical Incident → Reasoning → Validation → Human Escalation), routing to tools, and executing actions; **RAG architecture and memory retrieval** (hybrid search: TF-IDF + semantic reranking, ChromaDB, section-aware chunking, citation grounding, durable context layer); **tool and skill integrations** (external API calls, document intelligence, structured data queries); **prompt architecture** governing agent behavior consistently across all sessions. 93.6% precision, 87.9% F1, pass@3 88.3% across 1,240 test cases — evaluated using AgentProbe.

EDUCATION

M.S. Data Analytics Engineering George Mason University, Virginia May 2026

GPA: 3.87

B.Tech Computer Science & Engineering JNTU Anantapur 2017

SKILLS

Agentic Runtime & Orchestration: LangGraph, LangChain, LangSmith, Multi-Agent Orchestration, Agent Sessions, Reasoning Chains, Tool Calling, Function Calling, Durable State Management, HITL, Conditional Routing, Orchestration Runtime, Prompt Architecture, Context Window Management, Structured Outputs, Memory Retrieval, Ambient AI, Real-Time AI

LLM Integration & Evaluation: Multi-Provider LLM Integration, LLM Routing, OpenAI API, Anthropic API, Groq, Ollama, Low-Latency Inference, Cost Controls, LLM-as-Judge, AgentProbe, Eval-Driven Development, Regression Harnesses, Golden Datasets, Benchmarking, Hallucination Detection, Offline + Online Evaluation, Trace-Level Observability, Agent Observability, OpenTelemetry

RAG & Knowledge Systems: RAG Pipelines, Hybrid Search, Semantic Search, Reranking, ChromaDB, Qdrant, Vector Database, Embedding Pipelines, Semantic Chunking, Contextual Retrieval, Citation Grounding, Knowledge Stores, Document Intelligence, OCR

Engineering Foundation: Python (8+ yrs), FastAPI, TypeScript, Node.js, PostgreSQL, Redis, Docker, Kubernetes, AWS (S3, EC2, Lambda, Bedrock), Terraform, GitHub Actions CI/CD, Privacy-Preserving AI, PHI Compliance, HIPAA, SOC 2, Distributed Systems, Fault-Tolerant Systems